THE ARCHIVES OF DENTISTRY

SUCCESSOR TO
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New England Journal of Dentistry.

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THE ARCHIVES OF DENTISTRY

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Societies.

BLEACHING TEETH.

BY DR. K. M. FULLERTON, CEDAR FALLS, IOWA.

The bleaching of teeth has, from the first introduction of a positive method, been met with remarkable indifference and, at times, positive prejudice. Why this should be will remain a problem. The teeth that require bleaching belong mostly to a class condemned for all purposes except that of mastication. The process is generally only applicable to the six anterior teeth, while possibly an occasional bicuspid may be treated with advantage. When these anterior teeth, especially the incisors, are discolored, they are such a positive disfigurement that the operator has only the choice of evils —to bleach or to excise them, and insert an artificial crown. It would seem no difficult matter to come to a decision, or at least to determine to give the natural tooth a chance for future usefulness.

The fear of re-discoloration, or annoying labor, should not be taken into consideration. All operators are liable to meet with sudden discoloration in the regulating of teeth by the strangulation of the pulp at the apical foramen. When this occurs, it is one of the most humiliating of accidents, as it is one of the most annoying to patients. Discoloration is caused by decomposition, through a slow disintegration of the organic material and the deposit of carbonaceous matter. It therefore follows that the products producing color are not *necessarily* taken into the tubes by imbibition, though doubtless, to a limited extent, this is the case, but are produced by local degeneration through putrefactive processes. This change, though very slow in producing results, eventually gives to the tooth the bluish tinge, or to a tooth long affected by decomposed matter, the dirty, bluish-yellow. It is unnecessary to enter minutely into the more remote causes of discoloration, but we may summarize them as follows:

1st. When death of a tooth is caused by a blow, attacks of caries, too rapid pressure in regulating teeth, etc., the death and devitalization are followed by imbibition of coloring matter through the largest diameter of the tubules and local discoloration of the tube contents in the minuter anastomosing conduits. These changes may occur in teeth affected by caries, or without any external evidence of disease.

2d. The more aggravated cases, when this color has changed to a bluish-yellow, involving the entire structure of the dentine.

3d. Of the latter class, there may be a further subdivision, in which these are complicated with periosteal lesions which more or less interfere with efforts at restoration to original color.

The necessity of making some effort to restore the color of teeth changed by devitalization was apparent to dentists very early in the present century. The constant destruction of pulps with the imperfect methods of practice then prevailing, necessarily increased this unpleasant complication to such a degree that treatment of the anterior teeth became, so far as appearances were concerned, of no value whatever. Under the defective modes of treating pulp canals then prevailing, discoloration was sure to follow the filling of teeth. Any attempt to change color is necessarily dependent for success upon preliminary measures. Without thoroughness *here*, all subsequent efforts will fail. The early attempts at bleaching, before the settled practice of filling root canals was established, was not a success, and it must ever remain a failure unless the minuter structure of dentine be carefully considered.

It has been demonstrated by artificial injection, and still better by sudden congestions of the pulp, that coloring matter may be carried nearly to the final distribution of the minute ramifications of the tubuli. This is an important point, for, without this vascularity, bleaching would be impossible. With it, the possibility exists of extending the whitening process to the peripheral border of the dentine; or, in other words, to its union with the enamel on the crown, or the cementum on the roots.

The diameter of the tubuli is so minute, always decreasing in size until lost in final distribution, that any agent used must necessarily require considerable time before it can penetrate to the minute tubes, therefore you must not expect to bleach rapidly and meet with good success. The change, if any change be made at all, is simply on the walls of the canal, and cannot penetrate to any depth of tissue. If the discoloration is superficial, this mode will be effected, but not otherwise.

Color can be changed by several of the acids: notably, oxalic and nitric. The former destroys the color, and the latter changes dark-blue to a yellowish tinge; but as both of these are very destructive, they should never be used except in connection with an ant-acid. The first named will be found to give better results when used in connection with chlorinated lime.

Chlorine, free, or in some of its combinations, has been and is to-day the main reliance for bleaching, and that it is the most effectual has been demonstrated.

Failures have *usually* been the result of defective manipulation. It has great penetrating power, is a thorough bleacher, is readily applied, and if handled with care, will prove harmless.

The possibilities of chlorine were early understood in connection with the bleaching of teeth, but the result was not satisfactory for the reason named, and also from the fact that no practical mode had been devised for its use. To present free chlorine to a tooth was an impossibility, owing to its irritating character and necessarily superficial action; and no plan had been originated to free it from its compounds, hence, all teeth suitable for bleaching were condemned to remain a perpetual disfigurement. The first attempt to present chlorine free to a tooth was made by Dr. James Truman, in 1862.

Chlorine is liberated from chlorinated lime by all the acids, but more rapidly by some than by others.

It was found that, as rapid action was not desirable, those acids that affected this were not satisfactory. Tartaric acid was one of these. The conclusion arrived at was that a 50 per cent. acetic acid was the best, although later investigation seems to indicate to the contrary. The difficulty attending the use of chlorinated lime is due to the fact that a good article is rarely to be found.

Good chlorinated lime is in the form of a dry powder; when moist, it is worthless. It should have a strong odor of chlorine. A rough test can be made by adding to a solution of indigo, in a test tube, a small quantity of chlorinated lime; to this, add strong acid, and note the rapidity of change in color. If this is very slow, or not accomplished at all, the chlorinated lime is unfit for use, and should be discarded.

INSTRUMENTS.—These, though very simple, require special notice, for neglect in this particular will involve total failure. *No iron or steel instruments should be used in any connection with the agent employed in bleaching.* This must be impressed on the mind of every operator. The reason for this is, that the salts of iron formed discolor the teeth very rapidly. It would be preferable not to use any steel instruments at any stage of the operation, but this is difficult to avoid in the excavation of the cavity. Instruments can be made of hard wood that will serve the purpose, but ivory, platinum or gold can be used in place of this—either of the latter materials making efficient instruments.

Extreme care must be used not to produce any unnecessary irritation. The removal of all remains of decomposed pulp from the canal is of vital importance, but this must not be done in a rough, rapid, careless manner. It is of great importance that no inflammation of the periosteum should supervene, as that not only complicates the operation, but renders it more doubtful of success. The removal of the pulp should be followed by the usual treatment given to a tooth, and no attempt should be made to change the color of the tooth until all evidences of putrefaction have been removed, which will be manifested in the absence of the odor of decomposition.

If this preliminary process has been satisfactorily conducted, the next step will be that of filling the canal at its upper third. Gold is claimed to be the best material for this. The question may be asked, why fill the upper third? Because it is absolutely necessary for success that the root should be bleached as well as the crown. It must be remembered that the pulp chamber requires the same careful treatment as that given to the canal. It must be thoroughly cleansed of all debris to its fullest extent, and that, in the incisors and cuspid teeth, is almost to the enamel line of the cutting edge. Having proceeded thus far, the case is now prepared for the further process of bleaching.

The next point to be considered is the insertion of the material. Before this is attempted the canals and crown should be well washed with a solution of ether, borax, sodium bicarbonate, or ammonia, to remove fatty matter. It should then be well washed with distilled aqua. The tooth is then dried, the rubber dam having been applied at the beginning of the operation. There are several methods of bringing the acid used in connection with the lime. This apparently simple matter is really quite difficult. One process is, to saturate the entire canal and pulp chamber with the acid before inserting the chlorinated lime. Another is, to dip the instrument into the weak acid solution and then into the lime, and pack rapidly into the cavity; and still another is, to make a paste by the use of distilled aqua and pack this in the tooth, and then apply a stronger acid by means of cotton wrapped around the point used. There are difficulties attending all these modes. The point desired must be kept constantly in view—that of having acid sufficient and of proper strength to break up the compound and set free the chlorine used, and to preserve as much as possible of the latter for bleaching. Before commencing the packing, everything should be ready, so the cavity can be sealed at once.

Convenience of adaptation must govern the choice of the material used for closing the cavity. Gutta-percha, oxyphosphate or oxychloride of zinc may be used with good results, but the zinc preparations are harder to remove than the gutta-percha. After sealing the cavity, the tooth must be left for a day or two. On the return of the patient, remove all of the application, avoiding the use of steel instruments. Syringe out the canal with distilled aqua. If the bleaching has not gone far enough, a second application must be made, and this be repeated until a satisfactory result is obtained. The importance of using distilled water must be insisted upon. The reason for this must be apparent, for in many waters the minerals held in solution, especially those impregnated with iron, will defeat the desired object.

The immediate bleaching effect will be observed on the lower third of the tooth where the dentine is the thinnest. In the majority of cases this will be effected by one application. The greatest trouble will be found at the gingival border. Here the dentine is very thick, and it will be slow work, and in some cases end in failure, to restore normal color. The great objection made to this operation is, that the tooth will re-discolor, but if the subsequent operations are properly performed, this danger will be reduced to a minimum. The fact that dentine is permeated by pulp prolongations throughout the tissue, increases the difficulty of bleaching, and also increases the liability of a return of discoloration; but if the oxidation of the soft contents of the tubes has been properly effected, and then an agent used to fill the canal, and also act directly upon this microscopic tissue, there is but little reason to fear a return of discoloration. The operation, simple as it is, requires close attention to details and a clear comprehension of possible results.

The tooth having been restored to a good color, the next consideration is the proper filling to place in it. In this connection the before mentioned fact still remains an important factor, that the tubuli are still filled with decomposable matter. To allow this to remain without attention to future contingencies, must result in eventual failure. To effect any good results the antiseptic must not only operate in the main canal, but penetrate deeply into the minuter conduits. This quality is possessed in a remarkable degree by chloride of zinc, and maintains the same effect when combined with the oxide of zinc, forming the oxychloride of zinc. The canal and pulp chamber should be thoroughly filled with this paste, or, it is better to line the whole cavity with it, and then finish with the oxyphosphate, using gutta-percha at the cervical margin.

Chlorine acts as a bleaching agent by reason of its strong affinity for hydrogen. Vegetable and animal colors when brought in contact with chlorine in the presence of water, is seized upon by the chlorine, and the oxygen set free, oxidizes the color and destroys it. Chlorine in this case acting indirectly as an oxidizing agent.

If you wish to try to bleach more rapidly, a solution of oxalic acid is used to liberate the chlorine. Oxalic acid is more rapid than either tartaric or acetic.

Sulphurous acid is said to be a good bleaching agent, and acts by an entirely different method from chlorine. It is therefore of great interest from a chemical point of view.

As before mentioned, chlorine acts as a bleaching agent by reason of its strong affinity for hydrogen. Sulphurous acid, on the contrary, is a reducing agent by reason of its affinity for oxygen, in combining with which it becomes *sulphuric* acid.

On the chemical character, therefore, of the coloring matter depends the choice of the agent to be used.

Chlorine should be used when the color is an oxidizable compound, or rich in hydrogen; while sulphurous acid should be used more particularly in substances highly oxidized and capable of being reduced.

There are a great many different agents used for the bleaching of the teeth, but I will not consume any more of your valuable time, as I find that my paper is already quite lengthy.

Dr. W. N. Morrison is very proud of his son, who carried off the gold medal at the Missouri Dental College.

INCIDENTS AND ACCIDENTS OF OFFICE PRACTICE.[1]

BY H. H. KEITH, D.D.S., ST. LOUIS.

There are no more useful lessons than those contained in the incidents and accidents of office practice. If we do not communicate the knowledge gained, the event is limited to the individual. Not alone should we record our successes and apparent achievements, that we may stimulate the energy of the younger members of our profession, but as faithfully read the story of our failures.

In 1877, “S. A.,” a boy of ten years of age was presented with a mesial corner of the right superior central incisor broken in such a manner that the pulp, though not exposed, had died. The tooth was much discolored, abscessed, and very loose. A few days treatment sufficed to bring the tooth into a comfortable condition, when the boy's visits ceased. Some time elapsed: when he next came the tooth was elongated fully one-half the length of crown. The gums presented a most unfavorable appearance, and extraction was at once pronounced as the only proper treatment. At the earnest solicitation of the boy's mother this was deferred until the next day, and such treatment applied as the case seemed to indicate. Just here it may be well to say that exploration showed the root was not fully developed, the canal being quite large and funnel-shaped. So marked was the improvement the next day, that all idea of extraction was dismissed, and the root was finally filled with gutta-percha. A temporary filling of oxyphosphate was then introduced, and allowed to remain for two years. Then the contour was restored with gold. This filling was again replaced six years later with another of gold, which remained to within a short time ago, when a porcelain faced crown took its place. Deferring extraction to the next day has saved this tooth for thirteen years so far, with prospects of many years valuable service yet.

The second case is that of a right inferior second molar, a root filled with gutta-percha being allowed to fill pulp chamber, on which was placed a gold filling, February 20, 1878. In 1887 the gentleman complained of discomfort, but it was sometime before the cause was ascertained. The tooth had been split through its antero-posterior length, the fracture terminating nine-sixteenths of an inch below the point of the crown, on the lingual side. The fractured piece was removed, and the gum pressed out by means of gutta-percha, to give a better view of the remaining root. It was finally decided to attempt to restore the tooth by means of a band and crown. The fragment removed was used as a model from which dies were made, on which was struck a piece representing the lost part, having extensions sufficiently long to encircle the remains of the crown. This, when adjusted in position, was partly filled up on the inside with gutta-percha. A porcelain cusp crown was then arranged to antagonize the superior teeth. For a time everything seemed to go well. A little inflammation about the margin of the gum upon the lingual side instead of decreasing, suddenly grew worse, and pus was formed at the point of division of the roots. This finally yielded to treatment, and now the tooth is apparently in perfect health. The cause of this fracture appears to have been elasticity of the gutta-percha, under the pressure of the gold filling.

Case 3:—E. W., a boy of nineteen years of age, had broken a point off the right superior central incisor, not quite exposing the nerve, which subsequently died. The accident occurred some five years previous to his visit to me. The canal was found large and funnel-shaped, and was treated in the following manner: The lower portion was enlarged a trifle more than the diameter at the apex. A piece of lead was then introduced, and found to extend to the top by accurate measurement. In order to produce an accurate adaptation of the lead to the surrounding walls at the apex, the lead was reduced with fine sandpaper, the scratches of the sand being parallel to the long axis of the tooth. When the lead was forced into place, these fine ridges could be seen to be flattened when examined with a magnifying glass, and an adjustment continued in this manner, until the lead was found to close the apical foramen completely. The filling was completed with gutta-percha, and a porcelain crown was mounted upon the root. This has remained in a favorable condition up to the present time, about a year and a half.

Case 4, is that of a central incisor, pulp destroyed, canal filled, in which a Howe screw-post was used as an anchor to secure a large contour filling. Some time after, the tooth began to show a decidedly green discoloration, near the neck, which gradually extended throughout the crown. The filling was removed and replaced, however, using a screw of silver and platinum instead. I have here two specimens of roots in which the Howe post has been used, and have seen two other cases in the mouth, the same green stain appearing in all.

When the Howe post was put upon the market by the White Manufacturing Company, their agents refused to tell of what metal they were made, but gave the impression that they were some form of platinum and iridium alloy. They proved, however, to have been made of chrome steel. Besides the disagreeable discoloration of all these roots, I am inclined to the believe that the chrome salt formed, acts as a constant irritant to the peridental or dental membrane, and will result ultimately in the loss of the tooth.

Case 5:—In this case the left superior second bicuspid was devitalized and became discolored. The gentleman who was the lady's dentist at the time, desiring to improve the appearance of the tooth, removed the dentine extensively on the labial surface, and proceeded to fill with gold. When the tooth came into my hands for treatment, I found the part of the filling against the lingual wall well condensed, but that against the frail labial wall quite soft, and this portion of the filling had leaked, and the tooth was again discolored, showing that in order to avoid undue pressure on the thin enamel wall, insufficient force had been applied to condense the gold. Would it not have been better in this case, and in fact in all similar cases, to have sacrificed somewhat the appearance of the tooth and made a more permanent filling by the removal of all that portion of the enamel which was liable to fracture.

Case 6, is one of those mistakes in diagnosis which are liable to occur in almost any practice. Miss E. presented herself with every appearance of an abscessed right superior second molar, a large sac protruding into the mouth, opposite the palatine root. The tooth was so extremely loose and so sore that the patient would not allow it to be opened. The abscess sac was opened and syringed out, and two days later the soreness of the tooth had sufficiently subsided to permit the removal of the filling. Drilling toward the pulp chamber, a short distance, developed the fact that the tooth contained a living nerve. The result of this case showed that the abscess was caused by the lodgement of a fragment of a wooden toothpick between the first and second molar.

Another case, in my own mouth. The second left superior molar had for years stood alone, which facilitated a thorough cleansing upon all sides: I was therefore somewhat surprised at what appeared to be the development of a case of pyorrhœa alveolaris. The tooth continued sore, becoming looser, until its removal was a necessity. Neuralgia, and all the symptoms of a dying pulp had been present for three months. On extracting the tooth the nerve was found to be alive, and not much congested. The three roots were absorbed upon their inner surfaces. Exploration of the socket revealed the fact that a portion of the process enclosed by the three roots had been entirely absorbed. As the socket did not close in the usual time, I made an examination, and the probe revealed the presence of the missing wisdom tooth. The tooth has still continued to come down, but has not yet reached the gum line.

Another case in my own mouth is of interest: the result of wearing a wedge for three weeks between the first molar and the second bicuspid, on the right side. Some time after the tooth was filled, the first bicuspid became sensitive to heat and cold, and showed symptoms of peridental inflammation. Had a patient come to me describing the conditions of this tooth, I think I should have at once drilled into it, and applied the arsenic, but as it was in my own mouth I did nothing; and for fifteen months this tooth gave more or less trouble, but finally these disagreeable symptoms subsided, and the tooth is now apparently perfectly well.

DR. G. L. CURTIS, of Syracuse, has been acting as Dr. Garrettson's assistant in oral surgery this winter, in Philadelphia.

CAMPHO-PHENIQUE. [2]

BY J. W. DOWNEY, M.D., STATE CENTRE, IOWA.

Mr. President and Gentleman:

Campho-phenique is a germicide and antiseptic or nothing, therapeutically considered; and discussing its properties necessarily opens the entire subject of germicides and antiseptics, a subject fraught with peril to the writer or speaker, especially if he is not a practical chemist, pathologist, and microscopist.

Nothing in pathology is better established than the fact that certain microscopic germs cause disease, and no point in therapeutics is better known than the fact that a few drugs will, within the limit of safety, destroy these germs, and thus most effectually cure or prevent disease.

In deciding which germicide or antiseptic to use, the dentist should enquire, 1st, which is the most effectual; 2d, which is the safest; 3d, which is the most agreeable to the patient. To answer the first question we must inquire of the experimenter. Dr. Frank L. James, editor of the *St. Louis Medical and Surgical Journal*, a pathologist and microscopist of large experience, has determined, by a series of over eighty cultures carried on during the summer time, covering a period of two months, that campho-phenique, pure, is equal to 1 to 85 of bichloride of mercury, which is six times as strong as it can be used even on the unbroken skin, and about 25 times as strong as is considered safe on cut surfaces.

I have purposely omitted comparison with other drugs of this class, as the bichloride was by far the most effectual of any in general use before the introduction of campho-phenique.

If these figures are correct, they answer the first question. Certainly, if campho-phenique is from 6 to 25 times as effectual as a safe solution of bichloride of mercury, then it should have the preference in all cases where it is applicable. To the second point, which is the safest germicide, we all should be competent witnesses. The mercuric bichloride is known to be a virulent poison, and therefore ranks lowest in this respect, with carbolic acid closely following it. Campho-phenique is absolutely free from toxic or caustic properties. This I have had frequent opportunity to prove, and no doubt many gentlemen present have had a similar experience. Applied to the unbroken skin it produces no sensation whatever. On cut surfaces there is a slight burning sensation when first applied, followed by anæsthesia.

Being non-poisonous, non-irritant, campho-phenique ranks first as a safe germicide.

Now to the third point, which is the most agreeable. The brassy metallic taste of the bichloride is intolerable, the taste and smell of carbolic acid and creosote are disagreeable to most people, and the odor and meagre antiseptic properties of iodoform should banish it from the operating room. Campho-phenique has a pleasant odor and agreeable taste, this should establish its claim as the most agreeable germicide. I have yet to hear the first patient complain of its odor or taste.

From the foregoing data I am led to conclude—

1st. That when used pure and undiluted, campho-phenique is one of the most efficient and reliable germicides and antiseptics.

2d. Being non-poisonous and non-irritant, it is perfectly safe.

3d. It is the most agreeable to the patient of any drug of its class.

I am glad to know that I am not alone in these conclusions. Prof. J. Foster Flagg, writing on this subject in the July number of the *Cosmos*, said, “When it is known that it is a notable germicide, an efficient antiseptic, a non-irritant, a decided local anæsthetic, non-poisonous, insoluble in water or glycerine, does not discolor or stain, is possessed of an agreeable odor and no disagreeable taste, and maintaining an unchanged integrity, it will at once be recognized as wonderfully adapted to a large proportion of dento-pathological conditions, from sensitivity of dentine through the varying conditions of pulp irritation, pulp devitalization, pericemental irritation, alveolar abscess, and caries and necrosis of contiguous osseous structures, and that thus it must rank as one of the most, if not the most valuable polychrest which dentistry possesses.”

It seems to me that this endorsement from a teacher and author of such acknowledged ability as Dr. Flagg, ought to place campho-phenique in the armamentarium of every dentist in the land. And now a word on its special uses, and I am through.

First and foremost as a pulp canal dressing in the various pathological conditions, from recent devitalization to alveolar abscess. Here it will take the place of corrosive sublimate, carbolic acid, creosote, oil of cassia, oil of cloves, iodoform, or any germicide heretofore used, except peroxide of hydrogen. If thoroughly rubbed on the gum or injected with a hypodermic syringe, it acts efficiently as a local anæsthetic, not equal however to cocaine, but there are no constitutional effects following its use, and there is no danger of the tissues sloughing. It is quite efficient as an obtunder of sensitive dentine.

The very disagreeable ache which sometimes follows the extraction of abscessed teeth is almost instantly relieved by placing a pledget of absorbent cotton saturated with campho-phenique deep in the painful socket.

These are a few of the chief uses to which this new candidate for favor can be applied; others will suggest themselves to each practitioner. Before closing I want to mention its use for a condition which is not in the realm of dental pathology, but which is a source of annoyance to every dentist who uses plaster and hard water. I refer to the condition generally known as chapped hands. It is one of the numerous forms of eczema, and is greatly relieved by campho-phenique. I use the following formula:

℞ Campho-Phenique, ...

Oil of Cade, ... aa ʒi
Rose Cosmoline, ... ℥i
M. Sig.—Apply frequently.

Campho-phenique should never be mixed with water or glycerine. It will mix in all proportions with alcohol, ether, chloroform, and all fatty substances. In dentistry it will seldom be necessary to dilute it at all. Gentlemen, give it a trial, and when you have weighed it in the balance of experience and found it wanting, we will assist you in writing its fate upon the wall.

ETHER AS AN ANÆSTHETIC. [3]

BY DR. A. C. KELLOGG, DECORAH, IA.

For over a quarter of a century ether, as an anæsthetic, has stood at the front of all anæsthetics, as the safest, most reliable agent to use in all surgical operations. Being a faithful advocate of this time-tried friend, which has done so much for humanity, a brief description of its qualities and effects will constitute the theme of this paper.

Sulphuric ether is prepared by distilling alcohol with sulphuric acid. For many years after its first discovery the profession were not aware of its anæsthetic properties, but looked upon it as a mere chemical curiosity. Amusing incidents are related of many who inhaled it for the exhilarating and intoxicating effects it produced. But to the late Dr. Horace Wells is probably due the gratitude we truly feel for giving to the profession its true anæsthetic properties, in the painless performance of all surgical operations, no matter how severe. Since that date it is used almost exclusively in all the leading hospitals, medical and dental colleges throughout the land.

Very few deaths have been reported from its administration, and, indeed, if that proper care and knowledge of the agent be used, together with a pure article, and an intelligent understanding of the pathological condition of the patient, the death rate would sink to a minimum, and I doubt not that if a death should occur, after all these precautions, its true reason might find an explanation in some other cause.

If it is desired to anæsthetize a patient, the most important thing to consider is the possession of a pure article of *ether*. There are several reliable makes. Squibbs' sulphuric ether, for inhalation, being one of the most reliable for uniform purity and freedom from heavy oil of wine, acetic acid, fusel oil, sulphurous acid, or excess of water and alcohol. Every operator should acquire the knowledge of testing ether, if it contain any of these adulterations, using none for inhalation but the purest article that can be procured. Just here it might be well to state that pure sulphuric ether has a specific gravity of 0.725°, boils at 96° and has a density of vapor 2.586. This latter fact should be borne in mind, and when administered in the evening the lamp or light should be kept away from the inhaling apparatus and bottle, for ether is inflammable, several accidents being reported where this precaution was not observed.

To insure the best results from ether, it should not be inhaled after a full meal. Dr. Turnbull recommends a biscuit or cracker, and a glass of wine or a tablespoonful of brandy, half an hour before, always avoiding for several hours previously the annoyance of a full stomach. Serious complications and deaths have resulted from lumps and particles of indigested food becoming lodged in the trachea and glottis, from the act of vomiting, as ether, with many people, produces vomiting, and a recent meal is often reproduced.

The apparatus for administering ether is very simple, consisting in a towel or newspaper folded in cone form, with a moistened sponge at the apex to receive the fluid. During the first part of inhalation it is well to hold the cone a little distance from the patient's face, that the first few inhalations may be mixed with atmospheric air, otherwise an oppressed, smothered feeling may possess the patient. This feeling happily passes away in a few minutes, and the cone may be held close to the face, bringing the patient under its influence as soon as possible; better results are obtained, the after-effects pass away sooner, and there is less danger of nausea than when administered slowly, taking a long time to bring the patient under its influence.

It is well to observe that the temperature of the room be warm and well ventilated, avoiding all draughts. The patient should be in a recumbent position—better perfectly horizontal, all tight garments around the waist and throat should be loosened, allowing perfect freedom to the organs of respiration. With a finger on the pulse, an ear to the breathing and an eye to the patient, the operator is to judge when anæsthesia is complete.

The physiological action produced can be summed in a few words. Observation shows that the functions of the cerebrum are affected first; next, the anterior or motor centers soon fail to respond to mechanical irritation, yet the functions of the medulla-oblongata (the center of respiration) are performed. This is the proper stage to appreciate, for, if the inhalation be still further carried on, the sensory and finally the motor functions of the medulla-oblongata are involved, and death ensues from paralysis of the respiratory centers.

In conclusion, I must not fail to observe that ether has a peculiar and exciting effect on the genital organs, and a prudent operator will not fail to have a third party present throughout all the period of anæsthesia, otherwise his honor and reputation might be forever blasted by the emphatic assertions of some female laboring under the unhappy delusion of having been injured beyond reparation.

DISCUSSION OF DR. STODDARD'S PAPER: PORCELAIN FILLINGS. [4]

President Briggs:—Gentlemen, I think we all are paying more attention to porcelain fillings than we formerly did. Since 1883 I have referred to them in my lectures as one of the methods of preserving the teeth, and have used them in my practice. One point particularly interesting to me is the method Dr. Stoddard uses, of packing the clay into the plaster impression, biscuiting it, then removing the surrounding plaster and finishing the fusing. I presume it is because the carver I have employed does not do this that he fails to give me good results from irregular impressions. I imagine he tries to take them out while they are in the clay, and of course, cannot, if the shapes are peculiar.

Dr. Smith:—My method of using porcelains is so similar to what Dr. Stoddard has just presented that my remarks will be largely an endorsement of his paper. I do only the operative part; the laboratory part is done by my assistant, so I have only that part requiring the shaping of the cavity and taking the impression. I have a number of questions I want to ask Dr. Stoddard on working his furnace, but that hardly comes in to what you would call discussion. I like the method I have used, that is, taking an impression of the cavity, baking the enamel and setting in cement or gutta-percha. I have also ground them in, and, as Dr. Stoddard says, it is a very difficult thing to grind them in entirely. Even a very large cavity will seem very small when you get the porcelain between your fingers and attempt to grind it into place. I think it is a much better way to take the impression and bake the body and enamel it, as Dr. Stoddard has suggested. I would further say, Mr. President, that I am using the porcelains where we find large cavities in molars: for instance, dead teeth, where we have a compound cavity, either the mesial or distal surfaces in connection with the crown, and where amalgam is prohibited and the teeth too weak for gold. I find that when an impression is taken of the cavity, and the filling made as Dr. Stoddard says in his paper, and set in cement, that it makes a very nice-looking filling, and one that wears exceedingly well. I use the porcelain in that way a great deal and obtain from it success and satisfaction.

Dr. Taft:—There is but little I can say on the subject before us, from the fact that I have had no experience whatever in making porcelain fillings; although so frequently do cases present themselves in my practice, where porcelain tips and inlays would no doubt make not only as durable but more artistic fillings, by far, than gold or any of the plastics, that I feel encouraged to adopt this method after listening to the interesting paper of the evening, and upon examination of the specimens before us. In looking over the specimens I notice quite an appreciable difference in color between the inlay and the tooth itself, more so in some than in others. This may, of course, be due to the fact that possibly the inlays were placed in some of them previous to extraction. I do not yet quite understand how the doctor mixes his material so as to get the color of the inlay as nearly like that of the tooth as is possible, and should like to have him explain the point a little more thoroughly.

DR. STODDARD:—I neglected to say that it was impossible to match the color of these dry, dead teeth out of the mouth, but there is no difficulty at all in the mouth. You have a baked sample of your body, which you keep, and from which you select your color.

DR. BIGELOW:—Mr. President and gentlemen: I have never used any of the porcelain fillings myself, but several cases have come under my observation, and the greatest objection that occurred to me, at least in those cases, was the well-defined line of demarkation between the filling and the tooth itself; not but what the porcelain was good color, but it was the material it was set in. A gentleman once opened his mouth and showed me his teeth, and spoke of the great pleasure and comfort that he had taken since his teeth were filled in that way. The porcelain fillings were made for him by a dentist in New York City. To me they were very much more unsightly than gold, possibly because the material used in setting them was not a good match in color for the natural teeth. I think I may have seen one of the cases that Dr. Stoddard has spoken of in his paper. So far as the porcelain itself was concerned, it was a very good match for the tooth, but the line of demarkation was very distinct, almost as much so as if gold was used, though perhaps the strength would be greater. I don't know, perhaps Dr. Stoddard manufactures his own cements and gets his shades just right, thereby overcoming this objection.

DR. TAFT:—There is one other place, Mr. President, where it seems to me these porcelain tips or fillings may not be always practicable that may be illustrated by a case in hand: namely, that of a patient whose upper incisors upon examination were found to be filled with fine fractures, extending along the surface of enamel from the biting edge well up towards the margin of the gums. In the left superior central I found what seemed at first to be a very small proximal cavity, and started very carefully to excavate it from the palatal side, when the corner of the tooth soon afterwards chipped off, and in still further excavating,—hoping to fill with gold,—it continued, in a most aggravating manner, to chip away more and more. To get the smallest possible undercut or groove to retain the gold seemed an utter impossibility, and the longer and more carefully I worked, the more discouraging it became, until finally I was obliged to give up altogether the attempt to fill the tooth with gold and to replace the broken corner with oxyphosphate cement.

Now, here would have been an excellent opportunity for a porcelain tip, provided a man had the requisite skill to get sufficient anchorage for it without experiencing the same difficulty that I encountered in attempting to make a gold filling. I should like to ask Dr. Stoddard what his own experience has been with this class of teeth: if it is possible to adapt porcelain tips in such cases, and if so, how long they would be likely to remain. They are the most discouraging sort of teeth, I think, we have to deal with, but fortunately, cases as bad as the one just cited do not confront us very often.

Dr. Stoddard:—I should think that that was a case where it would be scarcely practicable to put in a porcelain filling, unless the tooth could be backed with platinum and the filling held that way, rather than by pins running into the tooth substance.

Dr. Allen:—Mr. President and gentlemen: I have had no practical experience with porcelain fillings, but I was much interested in the paper just read. While I was abroad last summer I met Mr. Dall, the gentleman referred to by Dr. Stoddard, and he showed me some very beautiful specimens of porcelain fillings in teeth which he had prepared out of the mouth. His method in dealing with proximate cavities in superior front teeth, where the lingual, proximal and labial edges are involved, is to build up the lingual, cervical, and half of the proximal walls with gold, leaving a cavity for the insertion of porcelain, which, when finished, is a great success from an artistic point of view, as it does away with the objectionable display of a large gold filling. Mr. Dall cuts his porcelain inlays from teeth manufactured by C. Ash & Sons.

Dr. Meriam:—Mr. President, the body referred to is Ash's Tube tooth body. Ash, I believe, has always refused to sell it in bulk. I know that a number of American gentlemen have wished to experiment with it. Either the Harwood or the Thompson blow-pipe will bake it; of course, it would be very easy to bake in the Stoddard furnace. I do not know that it has ever been imitated or reproduced in this country. Of course, we often hear of Dr. Herbst's glass fillings.

There is one question I would like to ask Dr. Stoddard, which he can answer after I have finished, and that is, how far his body corresponds with the body usually used for the porcelain teeth of the shops? Does it have to fuse at a lower heat, or is it substantially the same?

It seems to me that, going further than this, an effort should be made, before they are entirely lost, to preserve the old formulas that are in the hands of the older dentists. I believe that the most successful manufacturers to-day are manufacturing teeth from the formulas of these men, and if they are available they will be useful to add to our directory, that they may go on record. I think this is a very interesting study, and it is certainly carrying us back to the older days of dentistry in some ways.

DR. ALLEN:—I have in my pocket a tooth which Mr. Dall prepared. He takes one of Ash's inlay teeth, cuts a groove in it and cuts them off in sections.

DR. STODDARD:—In reply to Dr. Meriam's question, the body that we use is Dr. Daniel Harwood's, and is harder than the ordinary bodies. There has been some effort, I believe, to preserve the old formulas. Dr. Preston has presented his to the school, and Dr. Chandler has some which he is preserving.

DR. MERIAM:—I think, seeing this specimen that Dr. Allen has passed around, that some years ago in France a porcelain was made in the form of a pencil, so that the end could be ground and fitted to a cavity, and then cut off and polished. I also believe they use the long teeth used for continuous gum work.

DR. SHEPHERD:—I noticed that some of the specimens, especially a tip for a central, were a little larger than necessary. I would like to ask Dr. Stoddard if the porcelain could be dressed down to give the proper contour to the filling.

DR. STODDARD:—Yes, it can be ground down and polished; and when it is wet in the mouth it looks just as well as English porcelain.

DR. CLAPP:—I have had very little experience in inlay fillings, but I find that the process of grinding in the piece of artificial tooth, when that is used, can be considerably facilitated by cementing the piece of porcelain into the end of a small stick with cements the same as is used by lapidaries who cut precious stones. I would like to ask Dr. Stoddard about the Ash teeth, they being softer and more easily fused than the bodies that we have, would it be possible to take those teeth and pound them up the same as the body is made now, and then use them as the body?

Dr. Stoddard:—That never occurred to me, but I think very likely it could be done. The only question it seems to me is in the coloring material, whether it would bleach or not.

Dr. Meriam:—I think we should make a distinction between the Ash ordinary tooth and the Ash Tube tooth. The tube tooth fuses at a lower temperature, and in soldering this tube tooth I found that it would change color and once, for a man with teeth very yellow, I took advantage of that fact and secured a very good match.

Dr. Clapp:—I would like to mention a case that came under my observation a short time ago. It was in the right superior lateral, the mesial portion being turned outwardly a very little. I noticed that there was a slight defect extending a little above and below the enamel, but no decay. I think I examined this tooth two or three times before I discovered that there was a porcelain inlay at that point. On questioning the patient, I learned that it had been put in by Dr. Rollins, eight or nine years ago. I find diamond disks the best for grinding inlays.

Dr. Meriam:—I remember hearing of a dentist standing by the chair of Dr. Perry, in New York, and his showing an operation he had done of that kind; it is now probably seventeen or eighteen years ago. The dentist pointed out that there was a check in his tooth, when it was a tip Dr. Perry had put on.

PRESENTATION OF SPECIMENS.

Dr. Cooke:—I wish to present a piece of steel which was sent me by Dr. Wetzel, of Germany. He got it from Geneva. It is very thin, and is a first-rate thing to use for a matrix, &c., and for passing in between the teeth where something very thin is needed.

A method of casting a plaster model where you take a bite and desire to get a model very quickly: First, cast one side, turn the impression over, place a double piece of bibulous paper over the plaster that is to form the tail piece and cast the other side. It comes apart without any trouble, doing away with shellac and oil, and is done with one mix of plaster.

Dr. Clapp:—A bit of vaseline will accomplish the same thing.

Dr. Cooke:—I have never tried the vaseline. I have a rather interesting case, an extensive piece of bridge and crown work which the patient received some years ago from a firm who make a specialty of this work. The bridge on the right side had broken away, several abscesses had formed, and the condition of the mouth was far from satisfactory.

Dr. Smith:—Mr. President, right here, after Dr. Cooke's remarks in presenting this case of bridge-work, and knowing that he himself has performed some operations in bridge-work, I would like to ask if the result of his experience places him on record as universally condemning bridge-work.

Dr. Cooke:—Far from it. I simply presented that as a specimen of bridge-work as performed in the nineteenth century. It was done by a firm that makes bridge-work a specialty.

Dr. Meriam:—These instruments, I think, have never been made in America. I had them copied by Mr. Schmidt. I present first the set of groove cutters or chisels for molars and bicuspids. I think you will find them about page 122, Appendix to Ash's Catalogue, 1886. They are of the well-known chisel form, and I will send around with them two made on my own curve. I like the Whitten Approximal Trimmers very much as a trimmer and also as a scaler, but I wanted something with a little suggestion of Dr. Lord's added to Dr. Whitten's, and in these the blade is flat and passes easily between the teeth.

While I am on the subject, I will show some of Dr. Lord's excavators that I have had made quite small, smaller than he has made himself, and for a simple proximal cavity where only one instrument is to be used, I think they are admirable. Dr. Lord only orders them in hatchet forms. Here are some hoes that I directed Mr. Schmidt to make for me; some of them have been rubbed down thinner.

H. L. Upham, D.M.D.,

Editor Harvard Odontological Society.

Send subscriptions to the ARCHIVES to Dr. DeCoursey Lindsley, 321 N. Grand Avenue, St. Louis, Mo.

DISCUSSION OF DR. PARR'S PAPER: IMMEDIATE SEPARATION OF THE TEETH. [5]

DR. J. G. PALMER:—I do not think I have anything in particular to say concerning rapid separation. I have the separator that Dr. Parr describes in his paper, and I also have a set of Dr. Perry's. The latter is, perhaps, the most nearly universal in its application of any. I cannot say that I agree with the doctor in regard to separating so rapidly in all cases: in some cases it is probably advisable to do it, but I would rather follow the lead of Dr. Faught's paper, and go a little slow. I think going slowly in separating teeth is as useful as it is in some other cases cited in the paper.

DR. C. A. MEEKER:—I have used Dr. Parr's separator for three years, and prefer it to any of the others that I have used.

DR. PINNEY:—Sometimes, when you have a couple of nice little teeth to fill, and you want to get to work and get it done, and you know your patient is going to be so nervous for a week after, if you put rubber between the teeth, that you cannot treat them, it is a great pleasure then to put on your separator and slowly and gradually get those teeth apart, with just one little squeal, so that you can put in your two fillings. This little instrument is one of the best things we have: it does not work in all cases, but it will in nearly all. It should be used carefully; you should not move the teeth in a second, or in a minute, but work carefully with it and you will be surprised to find how many times you can use a separator to advantage. Patients are better satisfied to have their teeth separated in this way, than be compelled to wear rubber or tape, or something of that kind, in the mouth—they say those things tire them.

Dr. Faught:—I would like to say a word about that one little squeal that has been spoken of. I do not think we dentists quite appreciate what that one little squeal means. I think it means a feeling of fear and distrust in our patients for years afterwards. I believe also that when we avoid giving pain to a patient, or stop a slight pain, it pays in our dealings with the patient in future years. They never forget it, but we forget them because we see so many of them, but each patient remembers that somewhere in his or her mouth there was one little turn given that caused intense pain, and they remember that *you* did it. It always causes a feeling of dread when they sit in our chairs again for another operation. I believe that we should, more than we do, try to protect our patients from every annoyance, however slight, even at the expense of a little more time. There are many cases where the use of mechanical separators is unobjectionable, and when it facilitates such work. I wish to call your attention to a method of separating not my own. It is the use of a little piece of tape. I hand a little piece of tape to my patient with instructions how to use it, and in the course of four or five days the necessary space is obtained, and quite painlessly. When they return the mouth is in condition for the operation, and they do not seem to have hurt themselves, and *you* have not hurt them.

Dr. Pinney:—After you have gained the beautiful space by the use of tape or some other appliance, do you not have sore teeth. One good point in favor of the separator is, that after you get these teeth separated, the pressure of the instrument is so positive that there is no pain whatever during the operation of filling. But when you separate them with tape, the teeth are loose and sore, and a little pressure upon them causes intense pain. The pain is so much greater than that which caused the little squeal in immediate separation, that almost any one would prefer the squeal.

Dr. Faught:—I would suggest the use of a little cocaine just before the operation.

Dr. Jennison:—I do not like to disagree with my friend, Dr. Parr, because he has given us a great many valuable things. I have not had much experience in the use of separators, for the reason that I cannot get my patients to submit to them. The moment I put one on and begin to turn the screw, the patient exclaims: “Take that off; I cannot stand it,” and I am compelled to take it off. Then in the use of the separator, I am always fearful of crushing the enamel, which I think is a very important thing to consider. With regard to separating immediately, it would be very desirable if we could do it under all conditions, or even under some conditions. I would like to see a separator of some sort for immediate use, and I have no doubt Dr. Parr can devise one, having plates not made of metal, and which will do the work of separating teeth without possible injury to the enamel. That, to my mind, is a very important point, and I agree with the essayist of the evening most emphatically that it is best to go slow in this matter. It has been said that teeth are sometimes so situated that you cannot pass anything through between them for the purpose of gaining space. When I find teeth in that condition, I introduce a piece of rubber dam; you can always get that in, and leave it there until the next day, and then I am usually able to put in something else, such as a bit of wood, which I generally use. I take time to separate. After I get space, if the teeth are still sensitive, I fill with gutta-percha and leave it for a few days, and soon the soreness is all gone.

Dr. J. A. Osmun:—I think it always best to be perfectly frank with your patients. If you must inflict pain, tell them you are going to do it, and why, but that you will be as careful as you possibly can, and you will find they will stand an immense amount of pain, and be satisfied when you are through.

Dr. Addleburg:—If you speak to the patients and tell them the operations will be painful, but that by inflicting pain you can serve them better, in most cases they will allow you to apply the separator, and they say it is much preferable to the old way of separating. I have some patients who will not allow me to use it, but most of them prefer the separator. Ladies especially say they prefer it, rather than have something between their teeth, when going into society; they would rather bear the little pain than the annoyance. I have used it for two years. After the teeth are separated I find them to be obtunded to such an extent that there is little pain in excavating and filling.

Dr. Eaton:—I want to add a word in favor of quick separation. For the last four or five years I have followed the practice of quick separation. If the separator fits property, and you use it with care, it does not cause much pain. As soon as you find you are inflicting pain, stop and rest a minute or two, and then you will be able to gain a little more space. The separator should not extend up to, or impinge upon the gum. A little gutta-percha under the bows of the separator will tend to keep it out of the gum and will also steady the separator. In some cases, where it is very difficult to keep the separator on the teeth, the placing of a little gutta-percha on the bows will overcome the difficulty, and the separator will stay in place, you can gain all the space required, and then the teeth are held firmly while you perform the operation of filling. Another thing: I have a number of dead laterals in my patients' mouths which I cannot account for in any other way than by slow separation. The pulps were not exposed at the time of filling, and yet a few months afterwards I found them dead. I believe the blood vessels were strangulated by holding the teeth in one position for so long a time.

Dr. Palmer:—I fear I did not express myself very clearly in regard to the different separators. I have frequently used Dr. Perry's separators, and have taken them off and put on Dr. Parr's; and I want to say, that while I do not find Dr. Parr's separator universal, as claimed, it is more nearly so than any other: and I can do with it what I cannot do with any other. But I have failed to find any patients like those Dr. Pinney speaks of, who prefer the separator to slower methods of separating. Many ask me never to use one again. I have found best results from using pieces of rubber and keeping them there until sufficient space is gained. I think the difficulty in Dr. Eaton's case was that the teeth were not kept sufficiently solid: if they had been held firmly, the difficulty would not have occurred.

TEETH A COMBINATION OF CONES.

At the usual monthly meeting of the Manchester (Eng.) Odontological Society, on December 10th, Mr. W. A. Hooton showed a collection of bones and specimens of ancient implements and pottery recently discovered in a limestone cavern at Deepdale, near Buxton, including remains of a brown bear, Celtic ox, deer, wild boar, fox, sheep, horse, and other animals.

The skull of the bear, which was in fine preservation, was found imbedded in a mass of stalagmite more than a foot thick. The specimen was an old one, and the teeth had been subjected to very rough usage, being excessively worn down and many of the pulp cavities exposed. The canines had all been fractured and afterwards worn smooth, with the exception of the right upper, which was of full length and encircled by a band of erosion. There was no trace of the second premolars.

The skull of a Celtic ox (*bos longifrons*) showed portions of skin in a petrified condition still adherent, and there was also half the lower jaw of a calf.

In the clay were found portions of a stag's antlers of great size, somewhat softened by exposure to moisture.

Although no human bones have so far been met with, the signs of man's presence were conclusive, and that probably during the ancient British and Roman periods. One antler had been divided, and the tip smoothed and sharply-pointed; another was shaped, apparently for use as a spear head; and close at hand a small carved bone ornament, much blackened, and some bits of bronze were found.

We know that fires were made in the cave, for fragments of charcoal are preserved in the stalagmite.

The specimens of pottery are unfortunately much broken, but examples of Romano-British and also of Samia ware have been identified by Prof. Boyd-Dawkins, also pieces of hand-made pottery.

Dr. Shaw said that what had been exhibited by Mr. Hooton referred us back to that almost eternity of the past when the limestone was formed in which these caves are now found—a time before the appearance of vertebrate animals. And even when, inconceivable ages after it was formed, the limestone had risen from the shallow seas and became a part of the dry land, these caves must have been formed in it at a date so remote as to be almost incomprehensible to our mental grasp. And they had undoubtedly been, from a time of which we have no record down to clearly historical times, the homes of animals—man included. Many of these animals have, in only comparatively recent times, become extinct. In many of these caves, however, are to be found remains that show they have been the homes of animals now only to be found in hot climates, but were able to roam far north of their present habitations at that period when this island formed part of a great continent which was connected with Africa, and possessed an altogether different temperature than at present. The ruder kind of pottery to which Mr. Hooton referred are probably of Neolithic origin. In regard to the ornaments and better class of pottery found in these caves, it shows they have been at some time inhabited by a race greatly superior to the ancient riverdrift and cave-men, and their still later Neolithic inhabitants; and there can be no doubt they were the places in which the Celtic and Roman element sought refuge at the time of the fierce Saxon invasion.

Referring to the inferior jaw of a wild boar in which there was, at the extreme posterior portion, a fully-formed molar tooth which had not yet erupted, and consequently, had not any of its cusps in the slightest degree worn down by mastication, he (Dr. Shaw) said that he had read a paper some years since before the Manchester Microscopical Society, in which he had vaguely hinted at a theory which he had not since had time to work out, but which he would now distinctly state and leave it to younger men to consider. Mr. Hooton had also exhibited a most interesting specimen of a young, partly-formed and unerupted horse tooth in which also there had not been any wearing down of the cusps. He (Dr. Shaw) had several specimens of the same sort. Now, this molar horse tooth was in reality a combination of five teeth with projecting cones of various heights. As soon as the tooth appeared in the mouth these cones began to be worn down in mastication, and the tooth eventually presented a flat surface with alternate layers of enamel, dentine and cementum, so arranged that the occlusion formed a veritable mill for grinding the food. Although the molars of the bear and boar are not made up in the same way as those of the horse, the wearing down can be seen to have taken place in the teeth of the bear and the boar exhibited; and if gentlemen will kindly examine this unerupted tooth at the extreme posterior portion of the inferior jaw of the boar, it will be seen what this animal's molars are like when first formed, and before they are put to any use. *They are ummistakably made up of a great number of cones.* Therefore, it was his (Dr. Shaw's) opinion that, while in the primitive manner dentine and eventually teeth first appeared there were no signs of cones, when this form did, in the long process of time, make its appearance, it became a starting point from which has been derived, by a great variety of combinations, the forms of the teeth of the higher animals.

THE TACOMA DENTAL SOCIETY.

We are in receipt of a copy of the constitution and by-laws of the above Society, which indicates that the dentists of the new State of Washington are abreast of the times.

“The object of this Society shall be to cultivate the science and art of dentistry, to promote among dentists mutual improvement, social intercourse and good feeling, and to collectively represent and have cognizance of the common interests of the dental profession in our city.”

We take pleasure in giving to our readers the code of ethics of this new Society as being something new and original.

Code of Ethics.

The members of the Tacoma Dental Society in the fulfillment of their duties to the profession, to the public and to each other, declare and accept as binding the following code of ethics as embracing such principles of honor, fairness and gentlemanly bearing as every gentleman of honor and self-respect should most willingly adopt.

Article I.

Section 1. It is the duty of every dentist to maintain the honor, respectability and good name of the dental profession, and by a manly and dignified bearing, by studious habits and mental improvement, as well as by a conscientious earnestness in the employment of his skill for the welfare of mankind, to aim at securing a general recognition of the worthiness of the dental profession to rank among the honored and learned professions.

Sec. 2. He should so practice his profession that the community will esteem it above the art of a mere mechanican and above traffic wherein shrewdness and cunning are an essential part of the stock in trade.

Sec. 3. He should therefore regard it as unprofessional and beneath the dignity of his calling to offer the products of his skill in competition at fairs, or to make sale of his services as does the shop-keeper of his goods, or resort to public advertisement such as cards, hand-bills, posters, or signs calling attention to peculiar styles of business, lowness of price, special modes of operating, or to claim superiority over neighboring practitioners, to go from house to house soliciting or performing operations, or to do other similar acts; but nothing in the above shall be so construed as to prevent any member from inserting simply his or her name, occupation and place of business in the public prints, or giving notice in the same of his removal, absence from or return to business, or issuing appointment cards with his fee bill thereon.

Article II.

Section 1. It is the duty of a dentist to treat the members of his profession—not excluding those who are his competitors—with the honor of a gentleman and the honesty of a true man; and when he has occasion to examine the operations of a neighboring practitioner, to do so without criticism.

Sec. 2. And when called upon to counsel concerning the utility of any operation, it becomes him to excuse any perceived faults which may justly be excused, and to make no attempt to undermine the confidence of a patient in a reputable practitioner, or by under-bidding, attempt to secure patronage that might go to another dentist.

Sec. 3. He should esteem it enough for honorable rivalry that the patient of another practitioner should from voluntary preference seek his professional skill. In short, he should treat every professional brother as he would his own brother in the flesh engaged in the same calling, or as a father would a SOD whose success he would not hinder.

Article III.

Section 1. A dentist should make his own personal advancement in the literature and practice of his profession his chief aim, and be determined to win success on the ground of merit. He should always employ his best skill, and should endeavor to instruct his patients candidly, knowingly and conscientiously in relation to their welfare as connected with their teeth.

Sec. 2. He should consider it also his duty to regard the needs of the poor in rendering gratuitous service, or by making such operations as are needful to health and bodily comfort available at rates within their means, ever regarding the wide difference between benevolently reducing the price to the *known poor* and selfishly reducing the price to gain patronage that might go to another dentist.

Officers and Members.

F. P. Hicks, *President*; W. E. Burkhart, *Vice-President*; A. J. Gustaveson, *Treasurer*; A. McCulley, *Secretary*; E. G. Case, C. Van Winter, W. S. Conn, W. Chamberlain, —— Lamson, I. A. Chapman, J. R. Kennedy, L. Eaton, W. H. Johnson.

IOWA STATE DENTAL SOCIETY.

The Iowa State Dental Society will hold its twenty-eighth annual meeting at Dubuque, Iowa, May 6th, 7th, 8th and 9th, 1890. For purpose of observing a great clinic and hearing read papers by the most noted writers in dentistry, all are invited to come. The officers are: Dr. F. M. Shriver, President; Dr. C. J. Peterson, Vice-President; Geo. W. Miller, Secretary; Ben Price, Treasurer. Executive Committee: Jesse M. Ritchey, C. J. Peterson, C. Thomas. Membership Committee: J. S. Kulp, J. J. Littler, A. B. Cutler. Publication Committee: Geo. W. Miller, T. A. Hallett, J. B. Entrikim. Vice-President Peterson is Superintendent of Clinics.

ILLINOIS STATE DENTAL SOCIETY.

The twenty-sixth annual meeting of the Illinois State Dental Society will be held at Springfield, beginning Tuesday, May 13th, and continuing four days.

Garrett Newkirk,

Secretary.

MISSOURI STATE DENTAL ASSOCIATION.

Mexico, Mo., March 15th, 1890.

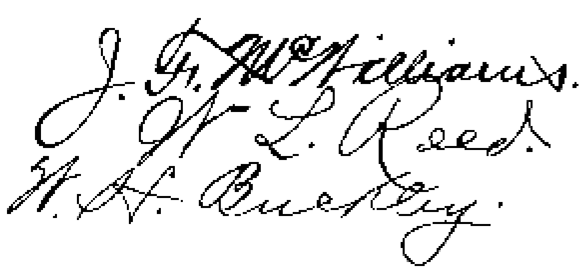
Dear Doctor: We wish to call your attention to the next meeting of the Missouri State Dental Association, which will be held at Pertle Springs, July 8–9–10–11, 1890.

No effort will be spared to make this meeting one of the largest and most interesting in the history of the Association.

The American Dental Association will meet in Missouri next August, and it is especially desirable that we have a large attendance at our next meeting so that we may make proper arrangements to receive the members of the American Dental Association in a manner that will reflect credit upon the dentists of Missouri.

Now is the time to make your plans so that you may be able to be with us, and we earnestly solicit your presence.

Fraternally yours.



J. H. McWilliams.
W. L. Reed.
N. H. Buckley.
Executive Committee.

KANSAS STATE DENTAL ASSOCIATION.

The nineteenth annual meeting of the Kansas State Dental Association will be held at Hotel Throop, Topeka, Kansas, commencing Tuesday, April 29th, and continuing four days. An interesting program is assured, and the clinics will be a special feature. The hotel is first-class in every particular, and has granted a reduced rate. One and one-third fare has been granted by the railroads, on the certificate plan. Members of the profession cordially invited.

C. E. Esterly,

Secretary.

Lawrence, Kansas.

FORTY-SEVENTH ANNUAL CONVENTION OF THE MISSISSIPPI VALLEY ASSOCIATION OF DENTAL SURGEONS.

CINCINNATI. OHIO.

The Society met at Lincoln Club Hall, and was opened by prayer by Dr. James Leslie. Dr. J. R. Callahan then delivered an address. He said—"As I look back over the records of the past doings of this Society, I can see where its founders builded better than they knew. I imagine this Society in its earliest days had more to do with bringing dentistry to its present high standing than most of us give her credit for doing. * * * In the very beginning they put the seal of their disapproval on charlatanism, amalgam fillings, advertising, derogatory remarks about one another in regard to poor ability, etc.; in fact it is hard to find anything that the most advanced of us to-day condemn, that they did not speak of in no uncertain tone forty-six years ago. Many are the good things done by this Society in early days; it has sown seed that has produced an hundred fold; many of the dental societies of to-day are offsprings of this old mother association. Not the least among the good fruits of this Society are the *Dental Register* and Ohio Dental College. * * *

“The Society spent money freely, both in the practical and theoretical of the profession; they gave prizes for papers to the value of $100; gave medals—gold and silver—for improved appliances. They seemed to be in great earnest in every way, and they did not forget to have good times, too, as they went along. From the records, I find that they were wont to gather about the festal board and break bread, crack chestnuts, and have a good social time, and at the close have what they chose to call interlocutory discussion. At one of the meetings, Dr. Somerby remarked that he thought it not in accordance with true pathological principles to retain a tooth in the mouth after the nerve had been destroyed, and that the operation of plugging over an exposed nerve, by capping or otherwise, would generally prove useless, and the idea of repeatedly tickling the nerve to make it cover itself with new bone, was more amusing than profitable. * * * *

“In the early years of this Society, it was truly *the* dental society of the Mississippi Valley. It drew its membership from all parts of the great valley, and often dentists were in attendance from over the Alleghenies. It was looked upon, and was truly *the* dental society of the West for many years; but in the forty-six years of its existence a new state of affairs has come about, in almost every State there are *local*, district societies, and State societies, and all auxiliary, more or less to the American Dental Society. Under this arrangement the Mississippi Valley Society is left somewhat isolated, and has lost much of its prestige. It has become somewhat local in its management, and it is with much difficulty that the programs are filled up each year. Men who write papers say, I have to attend my local society so often, I don't see how I can add another society to my already heavy burden, and many of the workers in dental society affairs are saying quietly, but with much significance, I wonder if the old Mississippi Valley Dental Society has not outlived her usefulness. As for myself I will not try to answer the question, it is a serious question and deserves thoughtful attention.”

DISCUSSION.

Dr. J. Taft said the paper brought many recollections of the past to mind, and was saddened when he found that nearly all the organizers had passed away, among whom were the Taylors, Griffith, Talbot and others. The work of this Society still has its influence upon modern dental practice. Much attention was paid to appliances and instruments, and many papers stand to-day, among which are those of Dr. Watt. The papers received more discussion at that time than at the present. The Society took the lead in all things.

Dr. James Leslie, Cincinnati, Ohio, spoke of the influence of the old Society, upon its future. The character of a society will be retained forever. Many of the important discoveries were made in this Society, and the ability of its members was equal to any in the world. The Society will live forever.

Dr. J. C. McKellops, St. Louis, said the Association could never die. Often there is a small attendance because it is not properly announced in the dental journals.

Prof. H. A. Smith, Cincinnati. Ohio:—This is the first Society he ever joined. The conditions are different now from what they were then, and so new methods must be used—new blood.

Prof. C. M. Wright, Cincinnati, thought the Society had passed its period of youth, and naturally, was entering upon its old age, and like a person looking in a mirror, he sees gray hairs which indicate a decline.

Prof. J. Taft stated that in 1855 the Society was at a very low ebb, and its continuation was obtained by the election of two new members at that moment.

Prof. J. S. Cassidy, Covington, Kentucky, said the discussion was premature (as it afterward proved to be), that the members have just begun to come in.

After some discussion by Drs. Taft and Callahan, it was decided to appoint special committees in conjunction with the executive committee to adopt new methods.

Paper by Dr. Otto Arnold, Columbus. Ohio:

NON-METALLIC PLASTIC MATERIALS FOR FILLING TEETH.

* * * "The employment of the earlier dentists of gums, mastic and sandarac, in etherial and alcoholic solutions for the stopping of cavities of decay, is the first approach history records of plastic fillings. About the year 1848, however, the first substantial progress was made in this direction by the use of gutta-percha as a temporary filling material. A little later, the well-known compound, Hill's stopping, was introduced, which is a modification of gutta-percha by the addition of certain mineral elements to make it harder, therefore more available for permanent fillings.

* * * About thirty years ago, oxychloride of zinc was introduced, the first of a now well-known class of filling materials, viz.: the zinc plastics. Next in order came oxyphosphate of zinc, followed by enumerable modifications and combinations.

* *

"It cannot be denied that the introduction of gutta-percha and the zinc plastics was the beginning of an era in operative dentistry that made it possible to attain results never before brought about. Prior to that time little, if anything, had been accomplished in the direction of protecting pulps from the effect of thermal irritation. The solution of this problem alone is of such intrinsic worth as to make any material, capable of contributing to that end, of inestimable value. All preparations of the zinc plastics, likewise gutta-percha, at least so far as the writer has knowledge, are more or less non-conductors of caloric, therefore valuable for this purpose, and it is almost an unpardonable offense to ignore their use in all large cavities as a protection to pulps. * * * Gutta-percha, however, unless in solution of chloroform or other volatile solvent, is not wholly safe, unless the greatest care is exercised to prevent its introduction into the cavity in too heated a condition. This is a serious obstacle, as the minimum degree of heat necessary to plasticity may, especially if the pulp is near the surface, be sufficient to permanently injure this organ. The pressure generally necessary to adapt this substance to place, is another objection. So, nothing short of the greatest caution in its use will give certain results. Gutta-percha as a filling material, compared with the zinc plastics for inside use, and amalgam for outer surfaces, has a limited sphere of usefulness. * * * The oxychloride cement has an escharotic action on organized tissue, which makes it unsafe as a nerve capping; but when used in connection with an intervening layer of a non-irritant, it becomes useful for this purpose. It is decidedly antiseptic, but readily soluble in oral fluids, and is distinguished as "the most preservative, and at the same time the most perishable of all filling materials." The antiseptic quality is a valuable feature for root fillings, and as these are supposed to be protected from the fluids of the mouth, their solubility is unimportant.

"The zinc phosphates are less irritating in their action on organized tissue, are denser in structure and less soluble in the oral fluids, and for general purposes are preferable and in more general use than the zinc chlorides.

"Briefly, then, to sum the matter up, what is the value of zinc plastics in dental practice, and to what extent should they be used? * * All large cavities should have a layer of this substance intervening between metallic fillings and their deeper portions, if possible, to protect the pulp from thermal irritation. * * * As a covering contiguous to exposed pulps, the more neutral and non-irritating of these preparations possess more good qualities than any other substance, chiefly on account of their adaption without pressure and the non-generation of heat.

"For filling root canals, zinc plastics are unsurpassed. The method I have practiced for a long time with more satisfactory results than any other, is to carry these to the apex on shreds of cotton of a fineness suitable to the case in hand, using necessarily the non-sticky variety. The facility and greater certainty with which the apex may be reached, combined with the imperviousness and antiseptic properties, make them the ideal root filling. For use in connection with crown and bridge-work, we have nothing to compare with them, and can only say they stand alone. For entire fillings in teeth that promise pathological complications, or for obvious reasons require temporary operations, they are a most valuable material. Taking them all in all, they occupy an important place in dentistry, and we could illy afford to return to the methods in vogue before their introduction. But like all good things, zinc plastics are often abused and their use is not always followed by the best results. * * I am opposed to temporary fillings as a substitute for something better, except possibly in children's cases, or where pathological or certain sexual conditions prohibit. The principal provocation for criticism is the indiscriminate practice of prostituting a good thing for uses other than its proper one. The outcome of such practice can result only against the general good of the profession, through the ultimate disappointment and loss to the innocent victim. The remedy that suggests itself against such abuse is to be more explicit in imparting advice on these matters. When temporary fillings must be inserted, impress the patient forcibly as to their limited utility. If such fillings are preferred on account of their inexpensiveness, or for any other reason, be emphatic in calling them temporary fillings and nothing more."

DISCUSSION.

Prof. J. Taft said he had never used amalgam as a filling material. Oxyphosphate acts differently in different hands. Had seen an oxyphosphate last eighteen years. It should not be used close to the gum, or in proximal positions. Is one of the very best materials for porcelain inlays. A *good* oxyphosphate will last as long as an *ordinary* gold filling. He uses it very often as an intermediate filling material. Much is due to the manner in which it is mixed. It should be thoroughly mixed. Some will granulate when used immediately, but if it is worked between the fingers, it becomes quite plastic and in that condition it is better and easier used. Heat accelerates, while cold retards setting.

Dr. J. R. Callahan asked if either the oxychloride or the oxyphosphate hardened the dentine.

Dr. J. S. Cassidy said oxychloride would harden by dehydration; much depends upon the healthy condition of the saliva. Mechanical abrasion has little to do with its loss. Chemical action is its chief cause. More loss takes place in mouths whose saliva is vitiated by putrefaction than by fermentation.

Prof. Taft said cleanliness is very important in mouths containing cement fillings.

Dr. H. J. McKellops spoke upon the oxyphosphates and their effect upon the dentine. If inserted in cavities and allowed to remain a year or so, when taken out the dentine will be found very hard and a gold filling may be inserted. It is one of the best materials for filling, and is especially applicable for children's teeth. Eructions from the stomach are very destructive to phosphate fillings. He says oxychloride is not the best root filling, especially in small roots. The mercury of amalgam will render a pulpless tooth very brittle.

PROF. H. A. SMITH:—Good manipulation is requisite to success in plastic fillings; the parts should be thoroughly combined. He had seen a large crown oxyphosphate filling which had been inserted eight years ago, in Germany. They were good filling materials, but had their place.

DR. L. E. CUSTER, Dayton. Ohio:—Before we can understand the action of the oral fluids upon the cement, or the action of the cement upon the tooth structure, we must first get an idea of what oxyphosphate of zinc is. When the two portions are united, the powder has such an influence upon the fluid as to cause it to crystallize, which in turn incorporates the powder in it. It is a chemical reaction so far as the liquid changes its form to a liquid of crystallization and also liberates heat, but it is a mixture as the powder does not change its form—it is still oxide of zinc. Being of this nature, it is easily acted upon by external agents, and at the same time the free phosphoric acid acts upon the tooth structure. He had found that alkalies acted upon the cement by neutralizing the acid of crystallization, which liberated the oxide of zinc as a precipitate. Strong acids overpower the phosphoric acid and act upon the oxide, the latter disappearing in the solution. So, as had been referred to by Dr. Cassidy, we will find a more rapid loss in mouths in which putrefaction is going on, because putrefaction being distinguished from fermentation only by the presence of nitrogen, and it has been shown by Dr. Watt that one stage in the formation of nitric acid was the formation of ammonia. Ammonia disintegrates the cement by neutralizing the acid of crystallization.

DR. ARNOLD said he filled roots with oxychloride by using fibres of cotton on a small broach.

DR. MCKELLOPS defied anyone to fill pulp canals with oxychloride of zinc, as well as with a solution of gutta-percha. He said there was often more than *one* opening at the apex. He can tell when the gutta-percha has reached the apex by watching the countenance of the patient.

Dr. Gray said he enlarged the root canal and filled with iodoform and oil of cassia.

Dr. F. A. Hunter, Cincinnati, Ohio, said he was never sure that he filled but two roots perfectly, and those were where the gutta-percha appeared at the opening of the sinus leading from the roots.

Dr. W. H. Sillito, Xenia, Ohio, asked if it was not the oxychloride of zinc which had the therapeutic effect.

The annual election of officers ensued with the following result: President. Dr. M. H. Fletcher, Cincinnati; First Vice-President, Dr. L. E. Custer, Dayton; Second Vice-President, Dr. Otto Arnold, Columbus; Corresponding Secretary, Dr. H. T. Matlack, Covington, Ky.; Recording Secretary, Dr. H. T. Smith, Cincinnati; Treasurer, Dr. Frank A. Hunter. Cincinnati.

(TO BE CONTINUED.)

Colleges.

MISSOURI DENTAL COLLEGE.

The commencement exercises of the Missouri Dental College were held in Memorial Hall, on the evening of March 13th, 1890, in connection with the St. Louis Medical College.

Prof. Alleyne, M.D., dean of the latter college, conferred the degree of M.D. upon twenty-two graduates.

Prof. W. H. Eames, D.D.S., conferred the degree of D.D.S. upon the following gentlemen:

William H. Auer, Jefferson City, Mo.
Thomas T. Baker, Litchfield, Ill.
Walter M. Bartlett, St. Louis, Mo.
Edward W. Bear, Sedalia, Mo.
Albert G. Bowman, Monroe, La.
Frank Henry Caughell, M.D., Morrison, Mo.
William A. M. Cumming, Farmer City, Ill.
John E. Deggendorf, Potosi, Mo.
Warden B. Dennis. Jr., Effingham, Ill.
Peter Henry Eisloeffel, 1102 Chouteau ave., St. Louis, Mo.
Henry D. Field, 1551 Lafayette ave., St. Louis, Mo.
John W. Forden, Springfield, Ill.
John J. Greer, Lexington, Mo.
Joseph Carter Goodrich, Wentzville, Mo.
Edwin C. Hammen, Jefferson City, Mo.

Guilford B. Housten, Carrollton, Mo.
Frank A. Kimler, Leroy, Ill.
Paul W. Keller, St. Louis, Mo. (College.)
Frank M. Lowry, Farmer City, Ill.
Marcus A. Mace, Belleville, Ill.
Peter H. Morrison, St. Louis, Mo.
Lorenz A. Naumann, St. Louis, Mo.
Charles W. Ott, Gardner, Kas.
Theodore L. Pepperling, Holstine, Mo.
Thomas N. Perrine, Anna, Ill.
Harry W. Pierce, Fort Wayne, Ind.
James H. Prothrow, Monroe, La.
Edward Schrantz, Warrenton, Mo.
Benjamin Q. Stevens, Hannibal, Mo.
David Riley Taggart, Campbell Hill, Ill.
Thomas E. Turner, Carrelton, Mo.
Edgar M. Whisett, Centerview, Mo.
Frederick V. Waldron, Evans City, Pa.
Francis W. Willard, Anna, Ill.

The prizes were awarded as follows:

THE ST. LOUIS DENTAL SOCIETY PRIZE, a Gold Medal, by Dr. Henry Fisher, to Peter H. Morrison, D.D.S., who received the highest vote on final examination.

THE J. WARREN WICK PRIZE, twenty-five dollars in gold, to James H. Prothrow, D.D.S., who received next to the highest Vote on final examination.

THE S. S. WHITE DENTAL MANUFACTURING COMPANY'S PRIZE, a set of Varney Pluggers, to Frederick V. Waldron, D.D.S., for excelling in operative dentistry.

THE ST. LOUIS DENTAL MANUFACTURING COMPANY'S PRIZE, a Laboratory Lathe, to James H. Prothrow, D.D.S., for the best specimens of artificial teeth.

A large and enthusiastic audience listened with pleasure to the music and speeches. Prof. John P. Bryson delivered the valedictory in behalf of the faculty.

The matriculants for the session were seventy-five, graduates, thirty-four.

DENTAL DEPARTMENT S. U. I.

The eighth annual commencement of the Dental Department of the State University of Iowa was held at the Opera House, Iowa City, on Monday evening, March 10, 1890. The annual address was delivered by Hon. R. G. Cousins, Tipton, Iowa. President Charles A. Schaeffer, Ph.D., conferred the degree of Doctor of Dental Surgery upon the following named gentlemen:

T. G. Albin, St. Louis, Mo.
J. V. Anderson, Cambridgeboro, Pa.
F. J. Bethel, Denver, Col.
A. D. Barker, Grinnell, Ia.
Benton Bement, Lockport, N. Y.
C. E. Booth, W. Superior, Wis.
C. M. Cobb, Clear Lake, Ia.
C. E. Coleman, Decorah, Ia.
G. W. Cook, Hyde Park, Ill.
Chas. Dorman, Manchester, Ia.
Andrew Dingwell, DeWitt, Ia.
J. H. Dorival. Caledonia, Minn.
F. E. Davoll, Madison, Dak.
J. W. Gluesing, Moline, Ill.
Nathaniel Glasgow, Maxwell, Ia.
C. H. Gibson, Chaska, Minn.
R. H. Guy Huntley, Mason City, Ia.
J. G. Hildebrand. Waterloo. Ia.
J. W. Hubbard, Muscatine, Ia.
Harriet Mabel Jones, Winterset, Ia.

W. H. Jallings, Washington, Minn.
Claude Kremer, Mabel, Minn.
F. B. Kremer, Caledonia, Minn.
R. E. Lamareaux, Ashland, Neb.
F. H. Low, Waukon, Ia.
W. B. Mandeville, Austin, Minn.
Edward Morton, Iowa Falls, Ia.
W. F. McDonald, Mt. Pleasant, Ia.
Chas. B. McCandless, Davenport, Ia.
W. E. Mabee, Sheldon, Ia.
G. C. Marlow, Lancaster, Wis.
E. H. Naumann, Oxford Junction, Ia.
H. O. Rogers. Ottumwa, Ia.
G. W. Schwartz, M.D.. Nebraska City, Neb.
S. L. Seeley, Manchester, Ia.
Richard Summa, St. Louis. Mo.
W. H. Simpson. Bellevue, Ia.
C. D. Tiffany, Mason City, Ia.
E. A. Taylor, Villisca, Ia.
P. L. Van Winter, Tacoma, Wash.
H. Van Winter, Marshalltown, Ia.
T. B. Wallace, Morrison, Ia.
Hattie E. Wells, Perry, Ia.

Matriculants for the session, 120.

Correspondence.

EDITOR ARCHIVES:

In the discussion of Dr. Faught's paper, reported in your March number, I am represented as saying: “The essayist remarked that the students who had graduated from a dental college did not know that it is the *acid* secretions from the mucous follicles which destroy oxyphosphate fillings.” I was quoting from a paper which I had heard read by Dr. Shepard, of Boston. I regret to say that I have misquoted him by inserting the word “*acid*.” His exact language, which I find in the printed report of his lecture, is, “Nearly every graduate seemed to be ignorant that there was any mucous follicles to contribute to the fluids in the mouth, etc.” He does not say “*acid*.” I have therefore misrepresented him, and wish to do him justice by this correction. In further explanation, let me say that I had heard Dr. Shepard's paper, and when at the Newark meeting, Dr. Faught said that *alkalies* destroy oxyphosphates, a gentleman next to me whispered, “Dr. Shepard said acids do,” upon which I seemed to recall that he had done so. I mentioned it, meaning to bring up the point that we can scarcely expect definite answers from students upon questions which have not been definitely decided. In another paragraph I use the expression “*wash out*,” in such a way that it sounds as though Dr. Shepard had used it. He did; but it was as a quotation from the students under criticism.

I hope that you will give this space.

Respectfully.

RODREGUES OTTOLENGUI.

115 Madison Av., N. Y.

Dr. Wm. Bartlett is demonstrator in charge at the Missouri Dental College. He has made mechanical dentistry a specialty for a number of years, but during the past two sessions has taken a regular course at the college, and is now a full-fledged dentist.

Editorial.

DR. JOHN ALLEN.

Some seventy-five dentists met at Sherry's Banquet Hall, Fifth avenue, on the evening of March 8th, to tender a complimentary dinner to John Allen, who has rounded the fiftieth mile-stone of active practice.

New York, Brooklyn, Boston and New Jersey were represented. All seemed delighted to do honor to the good doctor, who has reached the period when age, rather than blood, tells. It is a good custom to thus honor and cheer those who have been faithful workers in our ranks.

Four years ago they thus dined and honored Dr. John B. Rich, and it cheered and warmed his heart. Three years ago Dr. W. H. Dwinell was dined and wined, and he has been renewing his youth ever since. Two years ago Dr. W. H. Atkinson rounded up the fifty years, and they dined and wined him, and he is yet, we are glad to record, one of the boys. This year it came Dr. John Allen's turn to be wined and dined, and we all hope the good cheer and friendship manifested will cause him, too, to rejoice yet these many years.

John Allen gave to dentistry the most cleanly and beautiful denture ever put in human mouth, and he has given to humanity and the world a character and life equally as clean and beautiful. Wherever dentistry is known—known as it should be in its perfection—there continuous gum is known, and there goes with it, always, the name of John Allen.

What a sweet thought and consolation it is to think and know that you have done something for the world, and that it appreciates and thanks you; to feel that our lives have not been lived in vain, and though we may not have received all the dollars we wished for, yet, we are rich in blessings and the good-will of all. If so, then John Allen is rich, for all the world blesses him.

SOUTHERN DENTAL ASSOCIATION.

The Southern Dental Association will meet in Atlanta, Ga., on Tuesday, the 15th day of July next, at 10 o'clock a.m.

A cordial invitation is extended to the profession to meet with us on that occasion. Any cases in practice of unusual interest, I would be glad to have reported.

From the number of papers and clinics promised, and the character and high standing of the writers and operators, the meeting promises to be one of unusual interest. Let none stay away from this feast who can possibly attend.

JOHN C. STOREY, Pres. S. D. A.

AMENDED DENTAL LAW.—The Mississippi legislature, just prior to adjournment, Feb. 24th, so amended the dental law that, hereafter, persons beginning the practice of dentistry in Mississippi, must pass an examination by the board of examiners, whether they hold a diploma or not.

JULIEN W. RUSSELL, M.D.S., of Brooklyn, N. Y., called on the profession in our city recently. He was on his way to the Pacific Coast, combining business and pleasure. The business is introducing his alloy and copper amalgam; the main pleasure is a visit to his mother and sister in San Francisco, Cal.

DR. KIRK, of Philadelphia has entirely recovered from his very severe illness, and was heartily welcomed by the First District Society, before whom he read a paper at the March meeting. Dr. Kirk's essays are always so carefully prepared, that each one is a contribution to the literature of the profession which will live.

THE DENTAL PROTECTIVE ASSOCIATION.—This Association has a large membership and a vast amount of testimony in regard to crown and bridge-work, antedating all patents owned by the I. T. C. Co. The time will soon be ripe for an increase in membership fee. Those who are not members have but a brief period in which to get in on the ground-floor, for the small fee of $10. If you wish to be protected, send your name and $10 to Dr. J. N. Crouse, 2231 Prairie ave., Chicago, Ill.

Brief Mention.

Dr. Ottollengui says that his instruments, which he originated for implantation operations, “are just the thing” for necrosis.

Dr. J. W. Aikin is now located at 1032 Main St., Kansas City, Mo. The doctor is the longest and thinnest dentist in the Missouri State Association.

Dr. J. H. Kennerly, demonstrator in charge of the Infirmary of the Missouri Dental College for the past two sessions, is now located in Leadville, Col.

The Southern Dental Journal came to hand a few days after the 15th inst. Dr. H. H. Johnson, the new editor, starts out well. We wish him every success.

Dr. Garrettson is exhaustively revising his work on Oral Surgery. The next edition will have much new material, including a number of new illustrations.

The Minnesota State Dental Association will hold its seventh annual meeting in Minneapolis, Wednesday, Thursday and Friday. July 9, 10 and 11, 1890.

M. G. Jenison, Cor. Sec.

Dr. T. R. Ross, formerly of Cedar Rapids, Iowa, is now located at 3904 Indiana Ave., Chicago, Ill. The Iowa Association looses one of its most efficient and active members by the above change.

Periosteal Inflammation.—I have found, sometimes, after treating sore teeth for several days without giving relief, and where the trouble was somewhat obscure, and had probably been diagnosed as periosteal inflammation, or something of a kindred nature, that a pill of calomel, 2 grains, soda bi-carb., 3 grains, taken just before retiring at night, brought things all straight next day. Try it on your next patient with a sore tooth. It certainly holds good in malarial districts.

A. H. Hilzim.

Drs. A. G. Bowman and J. H. Prothrow have formed a partnership, for the purpose of practicing dentistry in Monroe, La. We commend them to the profession in Louisiana. Both are recent graduates of the Missouri Dental College.

Dr. C. W. Lewis, Chicago, Ill., on account of overwork and partly ill-health, has been taking a Southern trip, in order to recuperate, taking in Chattanooga, Lookout Mountain, Atlanta, Jacksonville, across the Gulf to New Orleans, then home.

The World Typewriter.—We are glad to call attention to the fact that at last a new typewriter has been invented that will fill the wants of the professional men. For further particulars, address the advertiser, H. M. Strader & Co., 608 Walnut St., St. Louis, Mo.

The Brooklyn Society has been “booming” things this winter. They have held public clinics once a month, and the attendance has been so gratifying that they have decided to continue them as a permanent feature of their Society work. This is a good idea. The First District Society of New York has become famous mainly in this direction. Other societies should try it.

Bridge-Work.—The most satisfactory form of sectional dentures (bridge-work) secured by crowns are made of gold and platinum, “I” or “L” bar, each end soldered to the crowns, and then a correct articulation obtained and rubber teeth used; the space between the crowns entirely filled with rubber, resting upon the gum, enveloping the gold and platinum bar, and articulating against the occluding teeth. I have used this form for many years.

Wm. N. Morrison.

DENTAL LAW.—There is a bill before the legislature of the State of New York, which provides that all diplomas granted to medical students, shall be conferred by a State board of examiners, the colleges being thus deprived of their present privilege. The board is to be appointed by the Regents of the State (who control educational interests in general), from candidates recommended by the State Medical Association. If this law is passed, it will be a step toward a similar action, in regard to the dental schools.

INTERNATIONAL MEDICAL CONGRESS.—*Rates*: the Hamburg-American Packet Co. will sell round-trip tickets by the Fast Line at a reduction of 10 per cent., and on the Mail Line at a reduction of 15 per cent., also the steamer August Victoria will take passengers on the going trip for a reduction of 25 per cent, on July 24th; the Wieland sailing on the 19th of July, will make a reduction of 15 per cent, on the going trip. For further information, address the company at 37 Broadway, N. Y., or P. O. Box 2567.

VICK—STATE FAIR, PEORIA, ILL., SEPTEMBER 29th, 1890.—James Vick, seedsman, of Rochester, N. Y., offers $1000.00 in cash premiums, to be awarded at the Illinois State Fair, by the Society's judges, for best cabbage, celery, potatoes, cauliflower, tomatoes, musk melon, onions, and mangel. Last year the prizes awarded at the New York State Fair went to Pennsylvania, Michigan, Iowa and New York. All interested in vegetables should send to Vick, of Rochester, for particulars regarding this offer. No doubt it will be one of the principal features of interest at the fair.

The Vicks will erect a separate building, or tent, in which they will make a grand show of flowers with the vegetables, and will be on hand to receive their friends.

HICKORY ROOT FILLING.—DR. WHITE protested against anyone condemning a thing of which he knew nothing. He had filled straight roots with hickory for eighteen or nineteen years. He does not fill the entire canal—not more than one-eighth of an inch at the apex. The object in using it, is to know that the foramen is closed; then you can fill the root with anything desired. The method is to file a piece of well-seasoned dense hickory almost to a point, then pass it up to the apex. If there is the slightest indication of pain, withdraw the wood, cut off a short piece from the end; again insert, mark at the cutting edge of the tooth: then again withdraw, and with a sharp knife, make a grove around it, about an eighth of an inch from the point, and bend the end over without breaking it off. Insert for the last time, the proper position being indicated by the groove, tap it home, and twist off the point.

Footnotes

[1] Read before the St. Louis Dental Society, Feb. 18, 1890.

[2] Read before the Eastern Iowa Dental Society, Jan. 14, 1890.

[3] Read before the Eastern Iowa Dental Society, January 14, 1890.

[4] Read before Harvard Odontological Society, December 26, 1889. ARCHIVES, Vol. VII, page 110.

[5] Read before the Central Dental Society of New Jersey. Archives, Vol. VII, page 49.

Transcriber's Notes.

1. Silently corrected simple spelling, grammar, and typographical errors.

2. Retained anachronistic and non-standard spellings as printed.

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